

Configurable Memory with CAM and SRAM

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Jiaxing Wang, Bryce Hwang

Introduction

Content addressable memory is an integral part of computing memory, providing efficient search functionality that is essential in its modern implementations, such as in databases, computer networking, and translation look-aside buffers (TLBs) in CPU cache.

There are two primary types of CAM. Binary CAM (BCAM) performs exact matching, where every bit of the search string must identically match the stored word to return a 1 indicating matching. Ternary CAM (TCAM) extends this by introducing the Don't Care bit (X), which matches both 0 and 1 in the search string. This allows a single stored TCAM entry to match multiple search strings, making TCAM particularly valuable in applications like IP routing and firewall rules where wildcard or prefix matching across ranges of addresses is required.

Despite the importance of CAM in modern computing, foundries have historically focused on density and performance optimizations on SRAM cells. Conventional BCAM requires 10 transistors per bit cell and TCAM requires 16 transistors, resulting in CAM arrays that are 2–5 times larger in area than an equivalent SRAM. This larger array area increases overall capacitance and power consumption, reducing energy efficiency per search operation. Furthermore, because foundries do not support push-rule layouts for CAM cells, the disadvantage of area compounds at advanced technology nodes where layout density is critical. As demand for high-performance, search-intensive applications continues to grow, bridging the density and energy-efficiency gap between CAM and SRAM has become an important challenge in Digital VLSI design.

Configurable Memory Theory

To address the area and energy inefficiency of conventional CAM, a configurable memory architecture is proposed that enables BCAM, TCAM, and SRAM operation within an array of

standard push-rule 6T SRAM cells. The 6T SRAM cell consists of two cross-coupled inverters that hold the stored bit, and two access transistors controlled by the word-line (WL). During standard SRAM operation, the word-line activates both access transistors simultaneously, connecting the storage nodes to the bit-line (BL) and bit-line-bar (BLB). The key modification enabling CAM functionality is splitting the single word-line into two independent lines: word-line-right (WLR), which controls the access transistor on the BL side, and word-line-left (WLL), which controls the access transistor on the BLB side. This creates two independently controllable access transistors without increasing area.

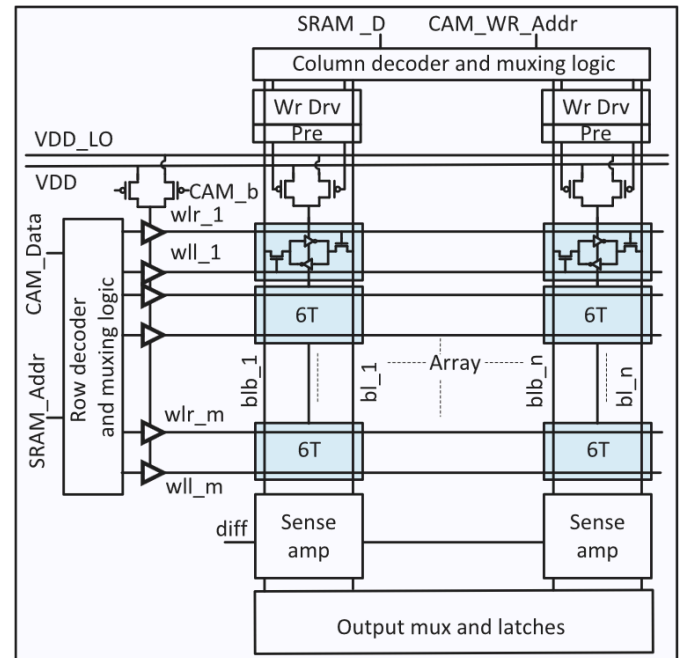


Figure 1. Configurable memory organization.

The fundamental architectural idea of the proposed design, shown in Figure 1, is storing words column-wise rather than row-wise. In a conventional CAM, words are stored row-wise, and the search string is applied on the bit-lines, with match lines running horizontally across each row. In the proposed design, each word is stored vertically

across a single column, and the search string is applied to the word lines instead. This reorganization is what allows the standard 6T cell to perform CAM matching without any additional transistors; the access transistors implicitly perform the bit-wise comparison through the charging and discharging behavior of BL and BLB.

During a BCAM search, BL and BLB are first precharged high. The search data is then applied to WLR and its complement, WLL. In a match, the stored bit and the search bit are identical on both sides of the cell, so neither BL nor BLB discharges — both sense amplifier outputs remain high, and their AND produces a match result of 1. In a mismatch, one or both bit-lines discharge below a reference voltage V_{ref} , causing the sense amplifier to latch to produce 0. This comparison occurs in parallel across all columns, preserving the parallel search capability of conventional CAM. Writing in BCAM mode requires two cycles due to the column-wise organization: the first cycle writes all 1s by enabling only the word-lines for rows requiring a 1, and the second cycle writes all 0s by applying data-bar to the word-lines. During both cycles, the word-lines are under-driven at a reduced voltage V_{dd_Lo} to prevent data corruption in neighboring columns, as discussed further in the **Design Considerations** section.

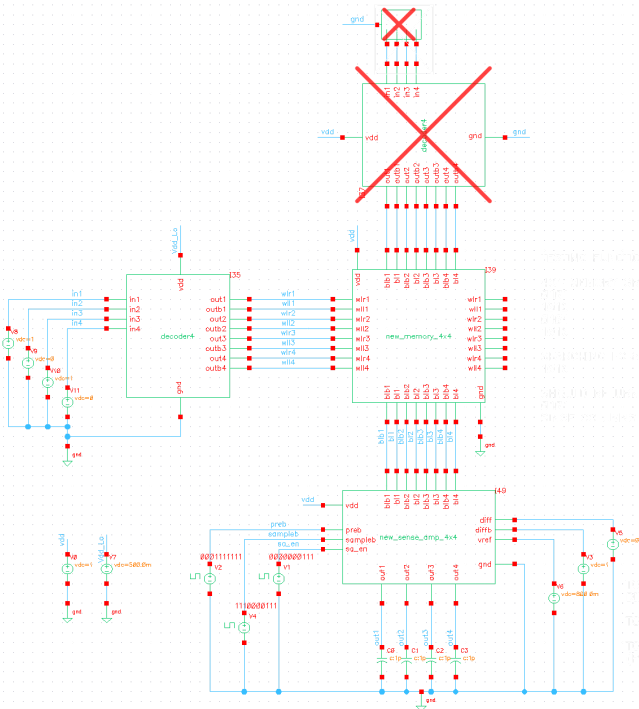
Figure 2. Memory design layout for BCAM search. The top decoder is ignored to test BCAM search.

TCAM operation extends from BCAM by supporting the don't-care state (X), which matches both 0 and 1 in the search string. Since a single 6T cell can only hold two states, TCAM uses two columns per word to encode three states: 0 is encoded as "00", 1 as "11", and X as "01". The don't-care encoding stores a 1 in both column positions, so neither BL nor BLB can discharge, regardless of the search string value: the bit always appears as a match. This behavior emerges naturally from the circuit without any additional logic. TCAM writes require three cycles: cycle 1 writes all "11"s, cycle 2 writes all "00"s, and cycle 3 writes the mask bits "01" for any don't-care positions. The tradeoff is that using two columns per word halves the storage capacity compared to BCAM mode.

Beyond CAM functionality, the configurable memory supports logic-in-memory operations by reusing the BCAM search circuitry to perform bitwise AND and NOR operations on rows stored in the array. This eliminates data movement between memory and the processor, reducing energy consumption. Rows to be operated on have their word-lines activated, while all other rows are masked by setting $WLR = WLL = 0$. The row decoder and muxing logic have control over each row's word-line drivers individually. For an AND operation, WLR is asserted for both target rows; if either row stores a 0 in a given column, BL discharges and the output is 0, otherwise it remains high, producing A AND B at the sense amplifier output. For a NOR operation, WLL is asserted for both target rows, reading the complement of each row onto BLB and producing A NOR B. Because only two word-lines are active during a logic operation, rather than all rows as in a full BCAM search, the search disturb risk is significantly reduced, allowing higher V_{dd_Lo} and faster operation compared to BCAM mode.

Configurable Sense Amplifier

A key component of the configurable memory is the configurable sense amplifier, which allows the same circuit to operate in two distinct modes depending on whether the array is configured as a CAM or SRAM. In a conventional differential sense amplifier, two cross-coupled inverter pairs work together to amplify the voltage



difference between BL and BLB. In this design, this cross-coupled structure is split into two independent parallel halves, controlled by a mode-select signal *diff*. This allows the sense amplifier to operate as two independent single-ended amplifiers in CAM mode or as a single combined differential amplifier in SRAM mode.

In CAM mode, $diff = 0$ and $diffb = 1$, meaning that the cross-coupled inverters operate independently. The upper half compares BLB against a reference voltage V_{ref} , producing output *outb*, while the lower half compares BL against V_{ref} producing output *out*. The timing sequence for each sensing cycle follows three phases. First, during precharge, $preb = 0$, charging the internal nodes of both halves to V_{dd} while BL and BLB are also precharged high. Second, during sampling, $sampleb = 0$, connecting BL and BLB to the sense amplifier inputs and allowing any mismatch to develop a voltage difference relative to V_{ref} .

Third, during evaluation, $sa_en = 1$, enabling the cross-coupled latches to resolve and output at the buffers. The outputs of the two single-ended amplifiers are then ANDed together; both must remain high to indicate a column match, while either falling to 0 indicates a mismatch. This two-single-ended approach is necessary because CAM mode must independently sense both BL and BLB to determine whether a mismatch occurred on either the 0 side or the 1 side of the stored bit.

In SRAM mode, $diff = 1$ and $diffb = 0$, tying the two cross-coupled halves together in parallel to form a single full-strength differential sense amplifier. Rather than comparing each bit line independently against V_{ref} , the combined amplifier directly senses the voltage difference between BL and BLB. This is faster than single-ended sensing because BL and BLB diverge simultaneously from the same precharged level, with one rising while the other falls, thereby allowing the differential signal to develop twice as quickly.

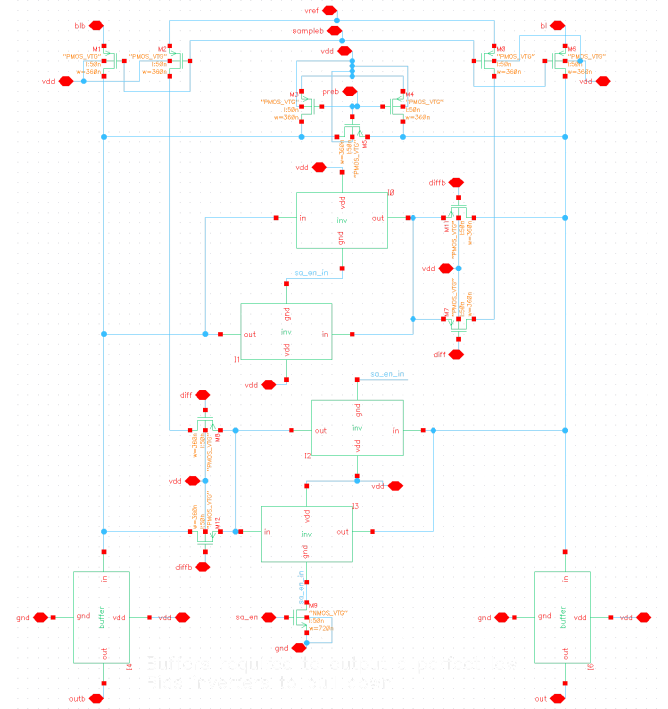


Figure 3. Configurable sense amplifier schematic.

Design Considerations

Several design considerations influence the reliability of the configurable memory design: transistor sizing and operating voltage.

Transistor sizing is critical to the proper functionality of memory cells and sense amplifiers, as a memory unit is sensitive to small voltage changes and may be corrupted by them. In the 6T SRAM, access transistors should be weaker than cross-coupled inverter NMOSes to prevent read upset. Given a minimum sizing of 90nm, we set the access transistors to 90nm while the cross-coupled inverter was set to be 120nm, providing an inverter-to-access sizing ratio of 4/3.

In the configurable sense amplifier, the access PMOSes (that access the bit line and bit line bar) are sized based on the number of cells connected to them. In our design in Figure 3, we have sized them to 360nm during testing to best accommodate a 4-bit line. The rest of the PMOSes are sized similarly to the sense amplifier inverters to be able to precharge, but not dominate them. This was selected to be 360nm as well.

Sense amplifiers inverters should have a low-skew since the sense amplifier is precharged to a high. When evaluating, transitioning to a low signal as quickly as possible is desired, hence a low-skew for our cross-coupled inverters is desired. The foot transistor of all four inverters, controlled

by sa_en , must also be sized correspondingly based on how strongly the sense amplifier inverters are skewed. For example, a beta of 1/2 (lo-skew) means that the foot transistor ratio must be 8 in differential mode. See Figure 4 below.

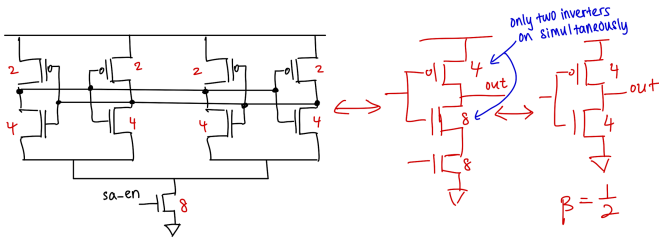


Figure 4. Equivalent transistor sizing calculation for the two cross-coupled inverters in parallel $diff = 1$ mode.

The buffers at the output of the sense amplifiers consist of two inverters. The second inverter should be no more than four times the size of the first inverter to ensure the second inverter is properly driven, while the first inverter should be no more than four times the size of the cross-coupled inverters to ensure they are properly driven.

Operating voltage is critical during BCAM operations. If many bit lines are used in BCAM search and BCAM write, many word lines will be activated, turning on more access transistors and thus accelerating capacitive discharge, which may unintentionally flip a logic signal and ultimately corrupt the data in memory. While the conventional solution is to increase access transistor sizing to combat BCAM data corruption, this change will also interfere with SRAM read upset margins in this configurable design. Thus, during BCAM operations, the word-lines and written-to cells are set to a lower voltage, V_{dd_Lo} , to complete their operations. This change reduces capacitive discharge significantly, eliminating the risk of data corruption.

Simulation and Results.

Our simulation was performed on a 4x4 memory array of standard 6T SRAM cells implemented in Cadence, shown in Figure 5. The array consists of four columns, each with a BL and BLB pair, four rows with independently controllable word-lines, and a configurable sense amplifier at the bottom of each column.

BCAM search and logic functions were tested, and the results match the proposed theory, shown in Figures 6-9. As for transition timing,

$t_{HL} = 1.27ns$ is recorded when $sampleb = 0$ during a BCAM search operation, meaning that a 4x4 array fully changes its logic state before the sense amplifier latches.

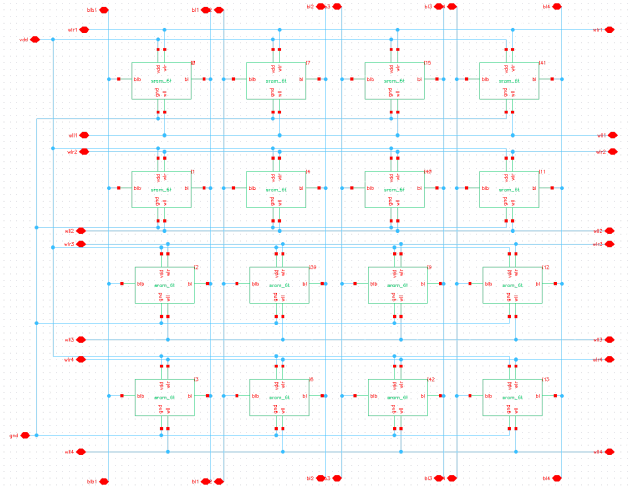


Figure 5. 4x4 memory array of 6T SRAM cells.

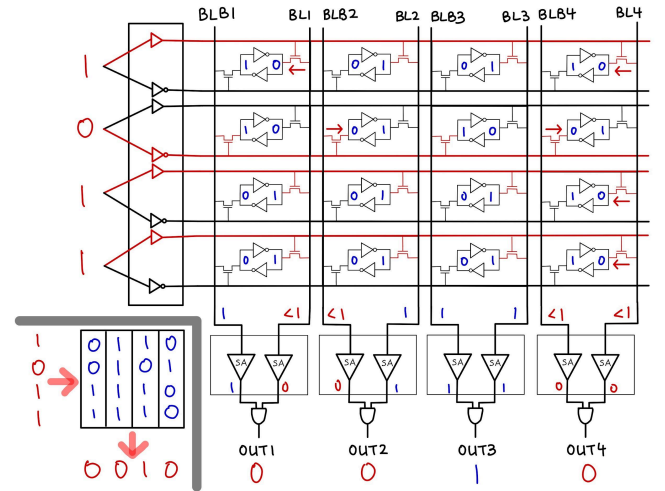


Figure 6. Example BCAM search operation. The configurable memory searches column-wise for 1011 and finds that column 3 holds the search value.

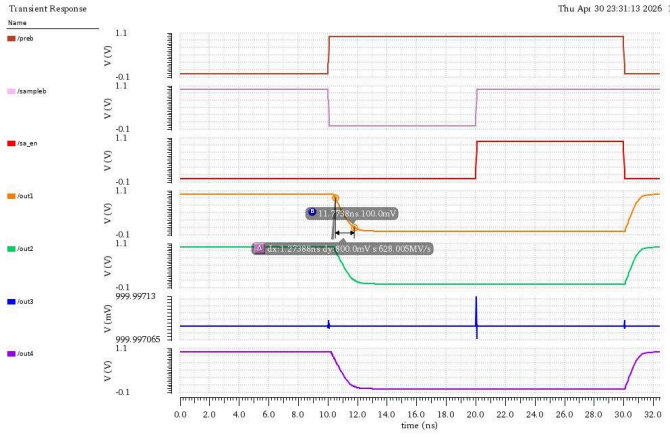


Figure 7. Transient response of the BCAM search operation shown in Figure 6.

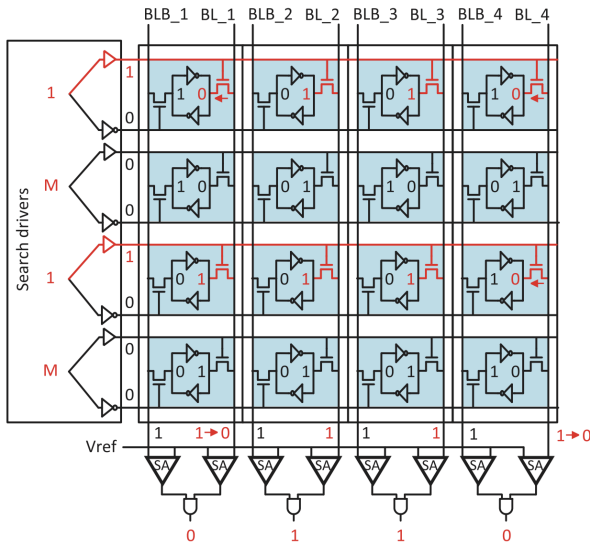


Figure 8. Example in-memory AND operation.

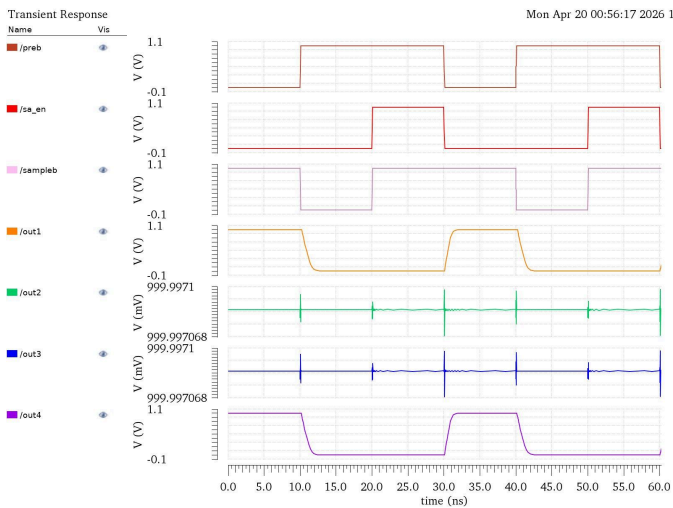


Figure 9. Transient response to the in-memory AND operation shown in Figure 8.

Conclusion

Our study of configurable CAM and SRAM finds that CAM density can be greatly improved by leveraging standard 6T SRAM cells, with minimal impact on SRAM performance. By storing words column-wise, splitting the word-line into independent WLR and WLL signals, and introducing a reconfigurable sense amplifier, the proposed architecture exceeds conventional CAM. The ability to switch between SRAM, BCAM, and TCAM modes on the fly further reduces the need for dedicated memory arrays, allowing chip designers to allocate memory resources dynamically based on the application's demands.

As transistor scaling slows at advanced technology nodes, improvements in circuit architecture and memory organization become increasingly important for achieving gains in performance and energy efficiency. The configurable memory demonstrates that substantial density and efficiency improvements are achievable not through new device technology, but through a clever reorganization of existing standard cells. This approach is particularly valuable in system-on-chip designs where area is at a premium and multiple memory types are needed on the same die.

The logic-in-memory capability introduced by this architecture points toward a broader shift in how computation and memory are organized in modern systems. As data-intensive workloads such as machine learning, network processing, and database operations become dominant, the energy cost of moving data between memory and the processor becomes a critical bottleneck. By enabling bitwise logic operations directly within the memory array without reading operands out, the configurable memory reduces data movement and offloads specific computations from the ALU. This positions configurable memory as a relevant building block for future near-memory and processing-in-memory architectures, where pushing computation closer to data is essential for continued gains in system energy efficiency and throughput.